

ENBC321/BIOE658I: Homework #6

Due: Monday 04/06/26 by 5:00 PM, beginning of the class

Submit your homework to ELMS, including Python code and the dataset

Overall points: 100

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: `problem1_fullname.ipynb`. Start with the problem number and followed by your full name, e.g., **problem1_jimmyazarnoosh**.

Note: Run all cells and ensure there are no syntax errors. All results must be displayed in the notebook. (Syntax errors = zero points).

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of ZERO. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- Python code files in ***.ipynb**
- Your answers to the questions in PDF format

Problem 1. Basic Metrics (50 points)

A hospital wants to develop a machine learning model to predict whether a patient has a certain disease based on medical test results. The dataset [medical_data.csv](#) contains anonymized patient records with the following features:

- **Age:** Age of the patient
- **Blood_Pressure:** Blood pressure reading
- **Cholesterol:** Cholesterol level
- **Glucose:** Blood glucose level
- **Diagnosis:** Ground truth label (1 = Disease present, 0 = No disease)

Your task is to train different classification models and compare their performance.

Tasks:

- Load the dataset into a pandas DataFrame.
- Split the dataset into a training set (80%) and a test set (20%).
- Train the following classification models: (10 points)
 - Logistic Regression
 - Random Forest
 - Support Vector Machine (SVM)
 - K-Nearest Neighbors
- Evaluate each model using: (10 points)
 - Confusion Matrix (True Positives, False Positives, etc.)
 - Accuracy
 - Precision
 - Recall
 - F1-score
- Compare the performance of different models. (5 points)

Conceptual Questions:

1. Which model performed the best, and why? (5 points)
2. If the goal is to avoid false negatives (FN) as much as possible, which metric should be prioritized? (5 points)
3. If we want a model that balances precision and recall, which metric should we focus on? (5 points)
4. Explain how overfitting and underfitting might affect the performance of your models. (5 points)
5. Which model is more likely to overfit: Logistic Regression or Random Forest, and why? (5 points)

Problem 2. Advanced Metrics (50 points)

This is a continuation of Problem 1, where we now evaluate the models using advanced classification metrics, specifically the AUC-ROC curve and the Cumulative Accuracy Profile (CAP) curve, to gain deeper insights into their performance.

Your task is to train different classification models and compare their performance.

Tasks:

- Load the dataset into a pandas DataFrame.
- Split the dataset into a training set (80%) and a test set (20%).
- Train the following classification models: (5 points)
 - Logistic Regression
 - Random Forest
 - Support Vector Machine (SVM)
 - K-Nearest Neighbors
- Evaluate each model using: (15 points)
 - AUC-ROC (Area Under the Receiver Operating Characteristic Curve)
 - CAP curve (Cumulative Accuracy Profile curve)
- Compare the performance of different models. (5 points)

Conceptual Questions:

1. Which model performed the best according to the AUC-ROC and CAP curve? (5 points)
2. Explain why the selected model is better based on these advanced metrics. In the context of healthcare, is a higher AUC-ROC always preferable? Why or why not? (5 points)
3. Describe what the AUC value tells us about the model's performance. What does an AUC value of 0.5, 0.7, and 1.0 mean in practical terms? (5 points)
4. Explain how the CAP curve can be useful in evaluating a model's performance for a disease prediction task. (5 points)
5. How does the CAP curve differ from the ROC curve, and in which situations would one be more useful than the other? (5 points)